

SRAP: Robust and Transferable Self-Reversible Adversarial Patch for Image Privacy Protection

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Abstract—Reversible adversarial examples offer adequate protection against malicious deep model identification and analysis. However, current methods still face challenges in terms of transferability and robustness, limiting their practical applicability. We introduce a novel technique for generating reversible adversarial examples utilizing Self-Reversible Adversarial Patch (SRAP) to address this. This approach significantly enhances the transferability and robustness of reversible adversarial examples against standard image processing techniques and adversarial defense methods. Specifically, we present a method for crafting adversarial patches that are small, non-overlapping, and adaptively integrated into specific regions. These adversarial patches are seamlessly combined with a reversible data-hiding technique that relies on prediction error expansion, resulting in adversarial examples with superior robustness and transferability. Experimental results indicate that our method achieves a remarkable transferability rate of up to 90% or higher between different models. Additionally, it exhibits strong robustness against image processing methods and adversarial defense strategies. Furthermore, our adversarial examples demonstrate an impressive attack success rate of 88% on commercial APIs, highlighting the effectiveness and practicality of our approach.

Index Terms—Reversible adversarial examples, adversarial patch, image protection, adversarial transferability, robust adversarial attack.

I. INTRODUCTION

RAPID advancement of artificial intelligence (AI) technology has raised significant concerns regarding potential privacy infringements. Several instances of unauthorized AI being exploited to analyze private information and steal data have been reported. For instance, some social media platforms leverage ambiguous user agreements to utilize user-uploaded image data for training their proprietary AI systems. Given the threat posed by such malicious AI, it is imperative to develop novel privacy protection methods that can effectively mitigate the risks associated with using AI technology.

Numerous studies have introduced adversarial examples generation techniques to mitigate malicious analysis and misuse of AI models [1], [2], [3], [4], [5], [6]. The core concept involves applying adversarial perturbations into samples that

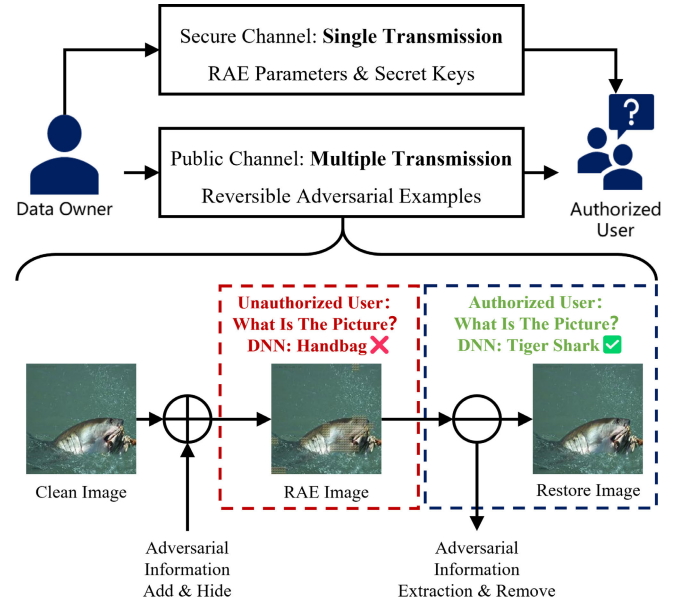


Fig. 1. In the framework of reversible adversarial examples, the data owner pre-distributes RAE parameters and secret keys to authorized users via a secure channel. While image data is transmitted through a public channel. This approach ensures that unauthorized users cannot analyze the image content using DNNs. Conversely, authorized users can utilize the provided keys to extract and eliminate the adversarial perturbations, thereby enabling the normal utilization of the image.

require protection, thereby impeding AI model analysis and ensuring privacy preservation. However these methods effectively shield data from analysis by malicious AI models but fail to restore the original data without additional information, hindering normal usage by legitimate AI.

To address this constraint, researchers have introduced Reversible Adversarial Example (RAE) generation techniques [7], [8], [9], [10], [11], [12]. As shown in Fig. 1, the core idea involves reversibly embedding the added perturbation information while generating adversarial examples. This method allows legitimate recipients to extract and remove the perturbations using a predetermined extraction algorithm or key, restoring the original sample.

Although existing RAE techniques have demonstrated promising results regarding attack success rate and recovery effectiveness, they still require further exploration in transferability and robustness. In practical scenarios, given the uncertainty of malicious AI models, relying solely on white-box attack success is inadequate for ensuring sample privacy

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protection. Furthermore, during transmission, samples are frequently subjected to various image processing techniques, such as lossy image compression. These processes significantly disrupt adversarial noise, leading to a loss of attack efficacy in RAE.

To tackle the aforementioned challenges, we introduce the robust and transferable **Self-Reversible Adversarial Patch (SRAP)** algorithm for image privacy protection. We propose an adversarial patch generation approach that relies on small, non-overlapping, and adaptively selected regions. We attain high attack efficiency with minimal information requirements by superimposing uniform, small-sized patches onto the sample's sensitive areas and optimizing their values iteratively. Furthermore, we employ Prediction Error Expansion Reversible Data Hiding (PEE-RDH) to quantify and embed the patch mask alongside the quantized patches within the adversarial sample, resulting in a highly transferable and robust self-recovering adversarial patch. The receiver can achieve nearly perfect restoration by extracting the mask and patch information using the same algorithm.

In summary, this paper makes two primary contributions.

1. We propose a robust and transferable method for generating SRAP. The method utilizes an algorithm that generates adversarial patches based on non-overlapping, small-sized, and adaptively selected positions. We successfully generate reversible adversarial patches by PEE-RDH.
2. SRAP outperforms existing approaches, achieving state-of-the-art attack success rate, recovery effectiveness and robustness against image processing and adversarial defense techniques.

II. RELATED WORK

This section provides an overview of the techniques utilizing adversarial examples and reversible adversarial examples.

A. Adversarial Patch Attack

Among adversarial attacks, adversarial patch attacks [13], [14], [15] apply intense perturbations to a localized image region. This design reduces the attack scope while endowing adversarial samples with significantly stronger transferability and robustness to real-world image transformations such as scaling, cropping, or rotation.

A defining feature of adversarial patches is the introduction of a spatial mask constraint, which strictly confines perturbations to the patch area and avoids altering non-patch pixels. Formally, adversarial patches are defined as:

$$x_{\text{adv}}^{\text{patch}} = x_{\text{clean}} + \delta_{\text{patch}} \odot M \quad (1)$$

where:

- $x_{\text{clean}} \in \mathbb{R}^{H \times W \times C}$ denotes the clean image, with H, W, C representing height, width, and channel count, respectively;
- δ_{patch} is the patch-specific perturbation term, allowing higher intensity due to its confinement to local regions;

- $M \in \{0, 1\}^{H \times W \times C}$ is a binary spatial mask: $M(i, j, c) = 1$ indicates pixel (i, j) in channel c is covered by the patch, while $M(i, j, c) = 0$ denotes non-patch regions;
- $\odot$ represents element-wise multiplication, enforcing $\delta_{\text{patch}} \odot (1 - M) = 0$ to keep non-patch regions unaltered and maintain global imperceptibility.

Foundational works highlight this localized, mask-constrained design: Ma et al. [13] leverages it for physical-world attacks, while Wei et al. [14] and Guesmi et al. [15] optimize patch structures to enhance transferability across models and resistance to image distortions.

B. Reversible Data Hiding

Reversible Data Hiding (RDH), a pivotal multimedia security technology since the 1990s, enables covert data embedding with lossless carrier recovery. Early RDH for uncompressed images established foundations through DE [16], HS [17], among others, while Sachnev et al. [18] developed Prediction-Error Expansion (PEE)—a dominant framework integrating DE and HS.

PEE utilizes pixel prediction errors (PEs) to generate Laplacian-distributed and sharply concentrated PEHs [19], achieving an embedding rate of 1.8 bpp [20] for efficient redundancy utilization. Its complexity is manageable: unlike MHM, which requires exhaustive parameter search [21], or Pairwise PEE, which involves intricate 2D mapping [22], it only entails PE classification and simple histogram shifting.

Notably, PEE has inherent robustness: prediction based on local pixel correlation enables it to absorb minor perturbations without impairing embedding and extraction, with extensions further boosting this tolerance. These merits make it ideal for reversible adversarial example tasks. Other adaptive methods (e.g., MHM [23] PVO [24]) have advanced RDH further, yet PEE stands out for its balanced performance.

C. Reversible Adversarial Example

Reversible adversarial example (RAE) technology emerges from the need to safeguard data against unauthorized AI use while ensuring legitimate AI can still access and process the data. This premise has shifted research focus towards embedding adversarial noise into examples without additional information transmission and extracting it with minimal loss. Existing methods primarily fall into two categories.

The first category employs RDH techniques to steganographically embed the noise [7], [8], [10], [11], [12], [25], [26]. The receiver can almost losslessly retrieve the noise and restore the clean sample using the corresponding extraction technique. This approach excels in maintaining high fidelity between the reconstructed and original samples.

The second category leverages a joint training framework consisting of a noise generator and a denoiser [2], [9], [27], [28]. The adversarial generator produces adversarial noise, which are then denoised to restore their original recognition results. This method accelerates adversarial example generation and avoids the computational burden of information hiding. However, it suffers from lower attack success rates and relies on the confidential sharing of the generator and denoiser,

compromising convenience. Moreover, the reconstructed samples exhibit significant differences from the originals, leading to inferior reconstruction quality.

This paper focuses on applying PEE-RDH techniques to embed and retrieve information within adversarial patches. Compared with RDH-RAE for global adversarial perturbations, SRAP adopts non-overlapping small-sized adversarial patches. By quantizing these patches, SRAP further reduces the required information hiding capacity, enhances the robustness of adversarial examples, and improves transferability via selecting key perturbation regions.

III. METHODOLOGY

This section details the methodology proposed in this paper, focusing on the generation of adversarial patches and the associated techniques for embedding and extracting adversarial information.

A. Problem Formulation

In data protection approaches utilizing reversible adversarial example techniques, two primary goals are pursued. Firstly, generating reversible adversarial examples is aimed at effectively compromising the target neural network's recognition capabilities. Secondly, attaining a high degree of recovery is essential, ensuring that, following restoration, the neural network model can accurately recognize the examples while maintaining good visual quality.

As shown in Equation (2), let x denote the sample requiring protection, with y being its correct label. For the target model f , it holds that $f(x) = y$. However, the adversarial example x_{ae} , generated by the adversarial attack method A targeting the model f , should induce a prediction error, resulting in $f(x_{ae}) \neq y$.

$$x_{ae} = A(f, x, y), f(x_{ae}) \neq y \quad (2)$$

As shown in Equation (3), for the generated x_{ae} , the adversarial information $(x_{ae} - x)$ should be embedded within it. We refer to this embedding process as E , resulting in the reversible adversarial example x_{rae} . This x_{rae} must have the ability to deceive the model f , causing it to make an incorrect prediction, i.e., $f(x_{rae}) \neq y$. Furthermore, for every embedding algorithm E , there must be a corresponding decoding process D , ensuring that the restored sample x_{re} satisfies two conditions: first, the model f should correctly predict it, second, the difference between x_{re} and the original x is less than a threshold δ .

$$\begin{aligned} \exists E, D, x_{rae} &= E(x_{ae}, x_{ae} - x), x_{re} = D(x_{rae}) \\ \text{s.t. } f(x_{rae}) &\neq y, f(x_{re}) = y, |x_{re} - x| \leq \delta \end{aligned} \quad (3)$$

To summarize, the objective of optimizing the reversible adversarial sample generation algorithm is to identify an appropriate combination of methods A , E , and D , ensuring that the protected sample x meets the previously stated criteria.

B. Overview

To address the aforementioned challenges, we introduce a method for generating reversible adversarial samples utilizing

SRAP. As depicted in Fig. 2, the process is divided into three primary modules: the attack method A : adversarial patch generation, which generates adversarial patches based on small, adaptively added, non-overlapping regions. The embedding method E , an encryption technique that employs prediction error expansion for reversible data hiding (PEE-RDH) to conceal adversarial information; and the extraction method D , which also leverages PEE-RDH to restore the original sample. Initially, an input image is processed using method A to create x_{ae} , which is then further processed by method E to yield $SRAP - x_{rae}$. Importantly, $SRAP - x_{rae}$ retains its adversarial capabilities, ensuring it can only be analyzed by authorized AI systems. For authorized users, the pre-established extraction method D can be applied to revert $SRAP - x_{rae}$ back to x_{re} . At this stage, the visual discrepancy between x_{re} and the original image x is indistinguishable to the human eye, and x_{re} remains fully recognizable and usable by AI models.

C. Adversarial Patch Generation

In the study of SRAP-based reversible adversarial example algorithms, several key challenges must be addressed. Firstly, the capacity limitation of reversible data hiding needs to be overcome. If the adversarial information exceeds the available capacity, it leads to information distortion, resulting in poor sample reconstruction and the persistence of adversarial patch information. Secondly, ensuring the visual fidelity of generated reversible adversarial examples is crucial. Our goal is to create examples that maintain adversarial effectiveness while remaining recognizable to the human eye. Lastly, robustness and transferability are essential attributes. The adversarial examples should exhibit significant resilience to common image processing techniques and adversarial detection methods, and demonstrate effective attack capabilities against unknown black-box target models.

1) *Gradient Mask Generation*: In SRAP, we utilize an adversarial patch generation algorithm characterized by small size, non-overlapping, and adaptive placement. Opting for a middle ground between global adversarial perturbations and adversarial patches, we employ smaller-sized patches. By strategically placing multiple identical small patches in critical regions of the image, we effectively induce adversarial attacks.

Hence, the placement of adversarial patches is of utmost importance. By strategically positioning these patches in highly sensitive and critical regions, we can ensure both the efficiency of their addition and a high attack success rate. Furthermore, this approach minimizes the likelihood of malicious users effectively removing the patches after data theft, thereby preserving the overall protection efficacy. This paper introduces a gradient-weighted method for selecting critical region masks. For an input image with shape of $H \times W$, given that the adversarial patch to be added is a B -side length square, the maximum number of patches that can be accommodated is $N_{max} = \lfloor H/B \rfloor \times \lfloor W/B \rfloor$. These potential patch positions are sequentially numbered from left to right and top to bottom, denoted as $[p_1, p_2, \dots, p_{N_{max}}]$, effectively partitioning the image into N_{max} grids. Subsequently, the target model is employed to process the input sample x , and the cross-entropy loss between the resulting logits and the true

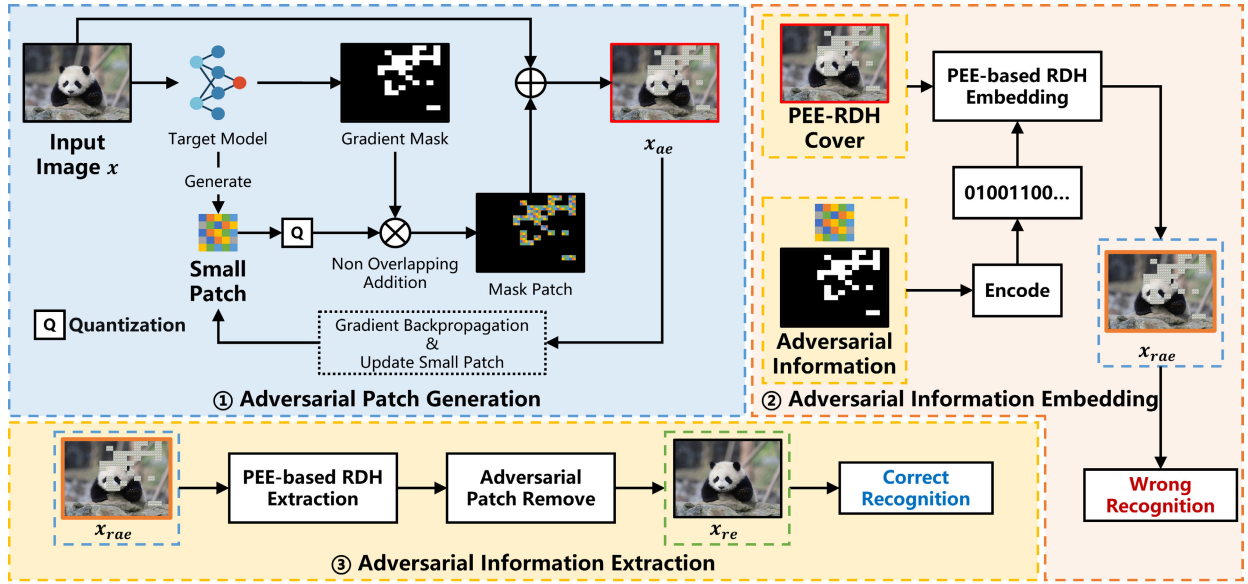


Fig. 2. The SRAP-based method for reversible adversarial sample generation and extraction comprises three key components: the generation of adversarial patches, the embedding of adversarial information, and the extraction of adversarial information.

label is determined. Additionally, as shown in Equation (4), the gradient g of the input sample concerning this loss is computed, with g 's size matching the image size and being similarly divided into N_{max} grids, $CE(\cdot)$ denotes cross-entropy loss function.

$$g = \nabla_x (CE(f(x), y)) \quad (4)$$

Let g_i represent the gradient block of the i_{th} candidate region. A larger sum of absolute values within this block suggests heightened model sensitivity to alterations in the corresponding image region, while a smaller sum indicates reduced sensitivity, the method for calculating the sensitivity of each block is presented in Equation (5). Consequently, by identifying the N most sensitive regions, we ascertain the optimal positions for placing adversarial patches in the attack process. Following this, the position information undergoes binary encoding. Specifically, in a sequence of length N_{max} , each p_i is encoded as 0 if not selected or as 1 if selected. Consequently, the mask is represented by a binary stream of N_{max} bits.

$$v_i = \sum_{m=1, n=1}^B |g_{i_{m,n}}| \quad (5)$$

In practical attack scenarios, we observe that even with a compact mask design, the required N_{max} length sequence for the mask may still surpass the steganographic capacity of certain images. To mitigate this, scaling the mask appropriately can notably decrease its informational demand while maintaining attack effectiveness. Thus, we adopt a fixed mask size of 15×15 in this study. Any chosen gradient mask with candidate positions is resized and binarized to this dimension, necessitating only 225 bits of storage. When applying perturbations, the mask is resized back to match the image shape.

It is acknowledged that this process may result in a misalignment between the perturbation positions in the mask

Algorithm 1 Gradient Mask Generation

Input: Target model f , input sample $x \in R^{C \times H \times W}$, true label y , patch size B , number of patches added N

Output: Gradient mask M

$N_{max} \leftarrow \lfloor H/B \rfloor \times \lfloor W/B \rfloor$

All candidate positions $P \leftarrow [p_1, p_2, \dots, p_{N_{max}}]$

Calculate gradient g via Eq. (4)

Calculate gradient patch value g_i via Eq. (5)

$M \leftarrow$ the N largest g_i

Resize M to 15×15

Resize M to x 's size

return M

and the initially selected candidate positions. To address this issue, we employ a non-overlapping superposition technique to expand the small-sized patch to the image dimensions before multiplying it with the mask to generate the new adversarial patch. Experimental findings indicate that this step has a negligible impact on attack performance but substantially reduces the storage requirement for adversarial information. In summary, the algorithm for calculating the gradient mask is shown in Algorithm 1.

2) *Small Patch Generation*: Once the attack mask areas have been determined based on gradient weights, it is necessary to introduce adversarial perturbations into each of these selected sub-areas. In this study, we opt for the utilization of adversarial patches. In comparison to global adversarial noise, adversarial patches inherently exhibit superior robustness and transferability [29], [30], [31], [32]. Nevertheless, our methodology diverges from prior adversarial patch attacks, which employed a singular, contiguous adversarial patch space. Instead, we generate smaller-sized adversarial patches and apply these identical patches to multiple chosen locations, thereby achieving the desired adversarial attack effect.

For a clean sample $x \in [0, 1]$ necessitating the creation of an adversarial patch, we consider a target model f such that $f(x) = y$, where y denotes the ground truth label of x . The image dimensions are defined as $H \times W$, with H representing the height and W the width. Initially, we must establish the size of the adversarial patch P as a $B \times B$ square and proceed with its initialization. In this context, Gaussian random noise is selected to initialize the perturbation, $P \sim N(0, 1)$, $B \ll H$ and $B \ll W$.

Adversarial Capacity Enhancement: when incorporating P into x , it is imperative to constrain the value range of P and apply it exclusively to the positions designated by the generated gradient mask. Nonetheless, a significant challenge emerges: conventional adversarial patch or noise generation algorithms impose a fixed range for adversarial perturbations, taking into account each pixel's range to ensure that $x + P$ remains within the $[0, 1]$ interval. However, when uniformly applying an adversarial patch to the mask-selected areas, the permissible addition varies across different locations. Specifically, if the pixel values in the i_{th} region span from 0.4 to 0.6, the allowed range for adversarial perturbations is $[-0.4, 0.4]$. Conversely, in the j_{th} selected region, where pixel values range from 0.1 to 0.9, the permissible range for adversarial perturbations narrows to $[-0.1, 0.1]$, markedly smaller than that of the i_{th} region. If this variation is not addressed, the overall range for adding adversarial perturbations across the image would be restricted to the narrowest regional range.

In the majority of images, selecting the smallest space for adding perturbations often results in a near-zero range for adversarial perturbations, thereby impeding the effective generation of adversarial patches. To mitigate this issue, it is necessary to scale the selected regions in the original image x prior to applying adversarial patches, ensuring sufficient space for their accommodation. Specifically, we first constrain the adversarial patch within the range of $[-\epsilon, \epsilon]$. Subsequently, for each selected region intended for patch application, we scale the pixel values around 0.5, yielding

$$x' = M \cdot (2(x - 0.5)(1 - \epsilon) + 1 - \epsilon) + (1 - M) \cdot x \quad (6)$$

This scaling ensures that the permissible range for adversarial perturbations in all selected regions aligns with $[-\epsilon, \epsilon]$.

The corresponding rescaling recovery method involves straightforward reverse calculations. However, this process incurs some image loss due to the discrepancy between the floating-point arithmetic used in calculations and the integer storage format within the $[0, 255]$ range during image transmission. As a result, rescaling introduces rounding errors. Nevertheless, our practical computational experiments have demonstrated that, within the current range of ϵ values ensuring attack effectiveness, the information loss is minimal. The impact on the final attack and recognition outcomes is negligible, validating the effectiveness of this rescaling method.

Following image rescaling, adversarial patches P are introduced. The identical patch P is applied to each candidate location within the selected gradient mask. During the iterative process, $x_{adv} = x' + P \odot M$ represents the sample augmented with the adversarial patch. Where $\odot$ denotes the operation of adding P to M at each position.

Adversarial Information Reduction: due to the constraints of current reversible information hiding methods, it is crucial to quantize the pixel values of the adversarial patch in order to reduce the necessary information capacity for its concealment. By adopting a quantization step size Q and limiting the adversarial patch's value range to ϵ , we designate the quantized patch as $P_q = \lfloor P/Q \rfloor$. In this scenario, P_q takes on integer values within the range of $[-\epsilon/Q, \epsilon/Q]$. For binary encoding, a maximum of $AP_{bit} = \lceil \log_2(\lceil 2 \times \epsilon/Q \rceil) \rceil$ bits are required to fully represent these values. After incorporating the quantization step, the process of adding an adversarial patch to an image to create an adversarial example x_{ae} can be obtained.

Adversarial Patch Optimization: Lastly, the small adversarial patch undergoes iterative optimization via gradient backpropagation to ensure a successful attack. The optimization procedure is outlined as follows. Initially, the adversarial patch is initialized using the standard normal distribution. This approach is chosen over zero initialization, which can cause gradients to vanish, and uniform random initialization, which may produce symmetrical adversarial noise and thus slow down the optimization. We denote the adversarial patch generated in the i_{th} iteration as P^i . Before applying it to a clean sample, the patch is quantized to decrease its information density as shown in Eq. (7). The $Clamp(x, -\epsilon, \epsilon)$ function represents a clamping function, which truncates the input tensor to a tensor with values ranging from $-\epsilon$ to ϵ .

$$P_q^i = Clamp(Q \cdot \lfloor P^i/Q \rfloor, -\epsilon, \epsilon) \quad (7)$$

Subsequently, the adversarial patch P_q^i is added in a non-overlapping manner to the regions selected by the gradient mask on the resized clean sample x' . The adversarial example generated in the i_{th} cycle is shown in Eq. (8).

$$x_{ae}^i = x' + P_q^i \odot M \quad (8)$$

In gradient-based adversarial patch optimization, the calculation follows Eq. (9). For the target neural network f , given the adversarial example x_{ae}^i generated in the i_{th} iteration and its ground truth label y , we initially compute the cross-entropy loss between the network's output for x_{ae}^i and y . Subsequently, we determine the gradient of this loss concerning the input adversarial perturbation P^i . The gradient information is then signed, multiplied by the update coefficient ϵ/Q , and added to the P^i from the i_{th} iteration, resulting in the adversarial patch for the P^{i+1} .

$$P^{i+1} = P^i + \frac{\epsilon}{Q} \cdot \text{sign}(\nabla_{P^i}(CE(f(x_{ae}^i), y))) \quad (9)$$

D. Adversarial Information Embedding and Extraction

After generating the adversarial patch, this paper utilizes a reversible information hiding technique grounded in prediction error expansion to enable the receiver to restore the clean sample exclusively from the adversarial sample, eliminating the necessity for supplementary information transmission. Specifically, this technique integrates the adversarial patch information and the patch location details into the adversarial sample. Consequently, upon receipt of the adversarial sample, the receiver can extract and eliminate the adversarial patch by

simply utilizing the pre-established information-hiding control parameters.

Adversarial Information Embedding: in an SRAP sample, the adversarial information comprises two components: adversarial patch information and patch location information.

The encoding of the adversarial patch information proceeds as follows. Initially, for a small $B \times B$ adversarial patch with three channels, quantization with a step size of Q allows each value to be represented by an $AP_{bit} = \lceil \log_2(\lceil 2 \times \epsilon / Q \rceil) \rceil$ binary string. Scanning each channel from left to right and top to bottom flattened the values into a one-dimensional array of length $3B^2$. A uniform offset of ϵ is added to ensure all elements lie between 0 and $2\lceil \epsilon / Q \rceil$. Subsequently, each value is encoded into a fixed AP_{bit} binary string. Ultimately, this process yields a binary string of length $3B^2AP_{bit}$, consisting solely of 0s and 1s.

Encoding the gradient mask for the adversarial patch location is relatively straightforward. Specifically, a 15×15 gradient mask is first flattened into a one-dimensional array by sequentially scanning from left to right and from top to bottom. This array is binarized, resulting in a fixed-length binary string of 225 bits. Subsequently, this binary string is concatenated with the encoded information from the adversarial patch, preparing the final information string $S = [s_1, s_2, \dots, s_k]$ for embedding.

In PEE-RDH, an adversarial sample, denoted as x_{ae} and augmented with adversarial patches, serves as the cover for embedding the adversarial information, regarded as the secret message S . Each channel is allocated to store one-third of the adversarial information for a color image comprising three channels. Each channel of the adversarial sample constitutes a single-channel image with dimensions $H \times W$. The predicted value for a pixel situated at position $x_{i,j}$ is determined by averaging the pixel values located directly above value $x_{i,j-1}$, to the left value $x_{i-1,j}$, to the right value $x_{i+1,j}$, and below value $x_{i,j+1}$. Subsequently, as shown in Eq. (10), the prediction error $pe_{i,j}$ for this pixel is computed as the difference between its actual pixel value and the rounded predicted value.

$$pe_{i,j} = x_{i,j} - \lfloor (x_{i,j-1} + x_{i-1,j} + x_{i+1,j} + x_{i,j+1}) / 4 \rfloor \quad (10)$$

In the cover, $3 \times H \times W$ prediction values are generated from the pixel values, followed by plotting a frequency histogram to represent these prediction errors. Prior research indicates that the distribution of prediction values closely resembles a Laplacian distribution [33], [34]. By designating the most frequent prediction error value as c , adversarial information can be embedded within the pixel positions corresponding to prediction errors of $a = c - 1$ and $b = c + 1$. Subsequently, during the information hiding process, the chosen pixel values denoted as $\{p_1, p_2, \dots, p_n\}$, and the corresponding prediction error is $\{pe_1, pe_2, \dots, pe_n\}$, then the obtained pixel values $\hat{x}_{i,j}$ are calculated shown in Eq. (11), where s denotes the embedding information bit.

$$\hat{x}_{i,j} = \begin{cases} x_{i,j} - 1, & pe_{i,j} < a \\ x_{i,j} - s, & pe_{i,j} = a \\ x_{i,j}, & a < pe_{i,j} < b \\ x_{i,j} + s, & pe_{i,j} = b \\ x_{i,j} + 1, & pe_{i,j} > b \end{cases} \quad (11)$$

Algorithm 2 Adversarial Information Embedding

Input: Adversarial example $x_{ae} \in [0, 255]^{H \times W}$, adversarial patch AP , gradient mask M

Output: Reversible adversarial example x_{rae}

$S_{ap} \leftarrow [s_1, s_2, \dots, s_{3B^2AP_{bit}}] = \text{bin}(AP)$

$S_{mask} \leftarrow [s_{3B^2AP_{bit}+1}, \dots, s_{3B^2AP_{bit}+255}] = \text{bin}(M)$

$S \leftarrow S_{ap} + S_{mask}$

$s \leftarrow s_1$

$c, a, b \leftarrow \text{PEEHistogram}(x_{ae})$

$x_{rae} \leftarrow x_{ae}$

for $x_{i,j}$ in x_{rae} **do**

 Calculate $pe_{i,j}$ via Eq. (10)

if $pe_{i,j} \in \{a, b\}$ **then**

 Embedding s via Eq. (11) into x_{rae}

$s \leftarrow$ next bit of S

end if

if S embedding completed **then**

 Exit the loop and terminate the embedding

end if

end for

return x_{rae}

Following the embedding process, specific pixel values within the adversarial sample carrier undergo subtle modifications, creating a reversible adversarial sample, denoted as x_{rae} . Despite these minimal changes, the adversarial characteristics of the sample remain largely preserved, enabling it to execute adversarial attacks effectively.

Adversarial Information Extraction: to extract adversarial information from x_{rap} , specific parameters must be predefined during the embedding process, precisely the size of the adversarial patch and the length of the adversarial information. The length of the adversarial information remains constant in the encoding and embedding phase. Upon receiving the reversible adversarial example, the initial step for the receiver is to utilize the prediction error outlined in Eq. (10) to construct a histogram of prediction error frequency distribution. Subsequently, they identify the maximum frequency value c and compute the spaces a and b , where the hidden adversarial information is located. The extraction process is reversible with the calculation method in Eq. (11), as shown in Eq. (12).

Traverse the pixel prediction errors with embedded information. If the value equals $a - 1$ or $b + 1$, the corresponding information bit s_i is 1; if the value is a or b , the bit s_i is 0. By examining all three channels, cease extraction once the length of the retrieved information aligns with the predetermined length. This method ensures lossless extraction of the adversarial information.

$$x_{i,j} = \begin{cases} \hat{x}_{i,j} + 1, & p\hat{e}_{i,j} < a \\ \hat{x}_{i,j}, & a \leq p\hat{e}_{i,j} \leq b \\ \hat{x}_{i,j} - 1, & p\hat{e}_{i,j} \geq b \end{cases} \quad (12)$$

Upon extracting the adversarial information S , it is divided into two components: the initial 225 bits signify the gradient mask, whereas the following sequence of $3B^2AP_{bit}$ length denotes the adversarial patch. Once separated, the relevant adversarial patch information is acquired. By employing a reversible information hiding algorithm to retrieve the carrier

Algorithm 3 Adversarial Information Extraction

Input: Reversible adversarial example x_{rae} , information length L

Output: Recover sample x

```

 $c \leftarrow PEEHistogram(x_{rae})$ 
 $a \leftarrow c - 1, b \leftarrow c + 1$ 
Adversarial information  $S \leftarrow []$ 
 $x \leftarrow x_{rae}$ 
while  $len(S) < L$  do
  for  $\hat{x}_{i,j}$  in  $x_{rae}$  do
    Calculate  $\hat{p}_{i,j}$  via Eq. (10)
    if  $\hat{p}_{i,j}$  in  $\{a - 1, b + 1\}$  then
       $S.append(1)$ 
    end if
    if  $\hat{p}_{i,j}$  in  $\{a, b\}$  then
       $S.append(0)$ 
    end if
  Restore value of  $x_{i,j}$  via Eq. (12)
end for
end while
Restore  $AP$  from  $S[: 3B^2AP_{bit}]$ 
Restore  $M$  from  $S[3B^2AP_{bit} : ]$ 
Restore sample  $x \leftarrow x - AP \odot M$ 
return  $x$ 

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x_{ae} and remove the adversarial patch, we obtain the scaled sample x' . Subsequently, rescaling x' allows for the recovery of the clean sample x .

To summarize, SRAP users can integrate adversarial information into adversarial examples utilizing a reversible information hiding technique, which involves adversarial patch generation and information embedding algorithms. These examples are then transmitted to authorized recipients via public channels. Upon receipt, the recipients can retrieve the original data for regular usage using a reversible information extraction algorithm. Furthermore, before embedding, the adversarial information can be encrypted using a key encryption method. This method ensures that even if unauthorized users gain knowledge of the reversible adversarial example algorithm, they cannot conventionally extract the information.

IV. EXPERIMENTS AND ANALYSIS

In this chapter, we will analyze SRAP's performance and compare it with the most advanced techniques currently available in the industry. Our comparison will encompass several key aspects, namely attack capability, transferability, robustness, and the efficacy of attacks on commercial models.

A. Experiment Setup

1) *Dataset*: In this study, we have chosen the annotated validation set of ImageNet2012¹ as our test dataset, as it remains independent of the target model training process. All images are standardized to 224×224 pixels with three channels.

¹<https://docs.pytorch.org/vision/0.20/datasets.html#image-classification>

TABLE I

THE IMPLEMENTATION DETAILS OF THE ATTACK METHOD PARAMETERS USED IN THIS PAPER

Adversarial Attacks	Attack Parameters
DP-RAE [7]	Super pixel size=4, unit perturbation=4/255, number of iterations for GQBE=10, PCT=50%, number of iterations for TISA=3000
BRAE [12]	PGN structure=Attention R2U-Net, training epoches=50, batch size=16, Adam learning rate=0.0001, range parameter=5, perturbation parameter=5
RAE-BIM [11]	Perturbation budget=16/255, number of iterations=50, super pixel size=16
RGAN [9]	Learning rate of G, D and R=0.001, training epoches=150, kernel size=2x2
GEAA [2]	Training epoches=200, batch size=128, Adam learning rate=0.1, MIN=10, T=0.9
SRAP (Ours)	Mask size=15x15, patch block size=5x5, number of iterations=50, perturbation budget=64/255, quantization step=8/255, patch number=256

2) *Target Models*: In this experiment, we chose target models from two categories: Convolutional Neural Networks (CNNs) and Transformer-based image classification models. Specifically, we selected AlexNet [35], Inception V3 [36], ResNet34 [37], ResNet50 [37], DenseNet121 [38], VGG16 [39], MobileNetV2 [40], and MobileNetV3-small [41] as the CNN-based target networks. We utilized Swin Transformer Tiny (swin-tiny) [42] and Vision Transformer Base 16 (vit-b-16) [43] for the Transformer-based models. All model parameters were obtained from the official pre-trained versions provided by torchvision V0.20.²

3) *Attack Methods*: In our comparison with the latest reversible adversarial attack methods, we focused on adversarial attacks utilizing reversible information hiding techniques, including DP-RAE [7], BRAE [12], and RAE-BIM [11], alongside attacks based on the generator-restorer architecture, such as RGAN [9] and GEAA [2]. Table I presents the attack parameters for each of these methods.

In the SRAP method proposed in this paper, the attack parameters are set as follows: a gradient mask with a 15×15 scaling size is employed, and the adversarial patches are 5×5 in size. The algorithm iterates 50 times, with a maximum perturbation amplitude of 64/255 added to each patch and a quantization step size of 8/255. Furthermore, 256 adversarial patches are added to each 224×224 image, covering 12.75% of the total image area.

4) *Metrics*: In assessing the performance of reversible adversarial examples, this paper employs three key metrics:

Attack Success Rate (ASR) quantifies the proportion of adversarial examples capable of misleading the target network after embedding adversarial information, as well as the misclassification rate of the restored samples.

The peak signal-to-noise ratio (**PSNR**) measures the discrepancy between the restored and original clean samples.

The structural similarity index (**SSIM**) is also adopted to evaluate the differences between the restored and clean samples.

²<https://docs.pytorch.org/vision/0.20/models.html#classification>

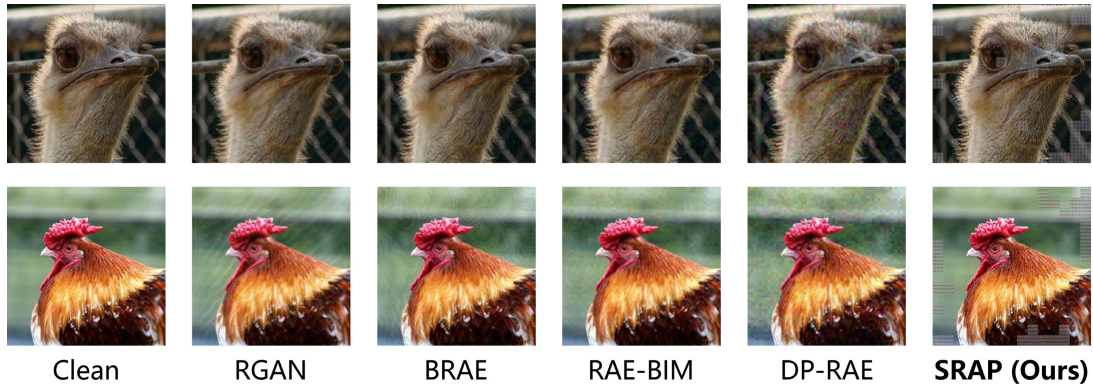


Fig. 3. Examples of adversarial samples generated by different reversible adversarial attack methods.

TABLE II

THE WHITE-BOX ATTACK SUCCESS RATES AGAINST THE TARGET MODEL ARE REPORTED, WITH THE OPTIMAL RESULTS EMPHASIZED IN BOLD

Attack Methods	Attack Success Rate in Target Models										
	AlexNet	InceptionV3	ResNet152	ResNet50	DenseNet121	VGG19	ModileNetV2	MobileNetV3-small	Swin-tiny	ViT-b-16	Average
DP-RAE	0.988	0.986	0.941	0.800	0.940	0.964	0.990	0.923	0.931	0.919	0.938
RGAN	0.905	0.992	0.986	0.736	0.985	0.897	0.967	0.980	0.922	0.988	0.936
BRAE	0.926	0.907	0.848	0.921	0.899	0.968	0.957	0.941	0.927	0.957	0.925
RAE-BIM	0.985	0.974	0.793	0.807	0.995	0.938	0.990	0.995	0.861	0.939	0.928
SRAP (Ours)	0.988	0.990	0.981	0.917	0.996	0.979	0.998	0.990	0.935	0.977	0.975

B. Attack Ability Evaluation

Table II summarizes the attack performance of different methods on ImageNet2012 target models. Since GEAA is a distribution-confusion-based attack operating on feature models, its effect on individual classifiers is difficult to quantify and is therefore excluded. Existing reversible adversarial attacks achieve strong white-box performance across diverse architectures. The proposed SRAP method attains optimal or near-optimal results on most models, achieving the highest success rates on DenseNet121, VGG19, MobileNetV2, and Swin-Tiny, including a peak rate of 0.998 on MobileNetV2. For models where SRAP is not the top performer, the performance gap remains marginal. For example, on InceptionV3, RGAN achieves 0.992 while SRAP reaches 0.990. Overall, SRAP achieves the highest average attack success rate of 0.975, demonstrating robust effectiveness across heterogeneous network architectures.

To verify that these gains are not caused by larger perturbations or longer optimization, all methods are evaluated under a fixed perturbation budget of 16/255. The attack success rate curves with respect to training or iteration cycles are plotted using InceptionV3 as the target model.

As shown in Fig. 4, generator-based methods improve gradually during training and converge at different efficiency levels, while gradient-based methods reach high success rates within a limited number of iterations and quickly stabilize. The proposed SRAP follows the same convergence pattern and achieves stable performance without prolonged iterations. These results confirm that the observed performance differences arise from optimization efficiency rather than extended computation.

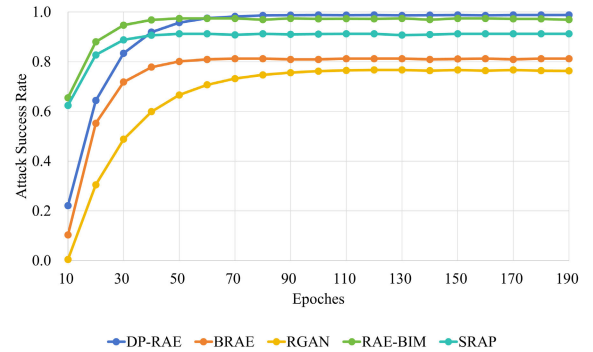


Fig. 4. Convergence curves of attack success rate with respect to iteration cycles for different adversarial example generation methods.

C. Adversarial Transferability Evaluation

In practical data protection scenarios against malicious AI, obtaining the structure and gradient information of the target network is often infeasible. Consequently, assessing the attack efficacy of generated reversible adversarial examples against unknown models becomes crucial. This paper presents an experimental comparison of adversarial example transferability. All examined adversarial attack methods involve generating adversarial examples on a target model and measuring the attack success rate on a test model.

Two strategies are considered for target model selection: utilizing a single target model, exemplified by InceptionV3 in methods like RGAN, RAE-BIM, and GEAA, or generating adversarial examples through an ensemble of multiple models to enhance transferability. Following the approaches of

TABLE III

FOR THE ADVERSARIAL TRANSFERABILITY EXPERIMENT, ALL ADVERSARIAL SAMPLES ARE GENERATED IN THE TARGET MODELS AND TESTED FOR ATTACK SUCCESS RATES IN THE TEST MODELS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Target Model	Attack Methods	Attack Success Rate in Test Models							
		AlexNet	ResNet50	DenseNet121	MobileNetV2	MobileNetV3-small	Swin-tiny	ViT-b-16	Average
InceptionV3	DP-RAE	0.327	0.268	0.474	0.460	0.286	0.352	0.410	0.368
	RGAN	0.309	0.457	0.411	0.548	0.198	0.395	0.511	0.404
	BRAE	0.243	0.571	0.587	0.392	0.330	0.388	0.447	0.422
	RAE-BIM	0.350	0.373	0.341	0.389	0.430	0.422	0.375	0.383
	GEAA	0.456	0.184	0.258	0.254	0.301	0.104	0.387	0.278
	SRAP (Ours)	0.869	0.535	0.697	0.748	0.889	0.495	0.699	0.705
InceptionV3+ ResNet152+ VGG19	DP-RAE	0.419	0.943	0.560	0.533	0.384	0.587	0.607	0.576
	RGAN	0.491	0.563	0.649	0.581	0.326	0.451	0.477	0.505
	BRAE	0.277	0.604	0.600	0.401	0.421	0.490	0.526	0.474
	RAE-BIM	0.368	0.404	0.417	0.450	0.511	0.455	0.440	0.435
	SRAP (Ours)	0.871	0.968	0.758	0.828	0.901	0.594	0.722	0.806

DP-RAE and BRAE, we select InceptionV3, ResNet152, and VGG19 for ensemble model attacks. GEAA, however, is not included in the comparison due to its incompatibility with ensemble integration. The experimental results of adversarial example transferability are presented in Table III.

The experimental results indicate that SRAP exhibits superior performance in most transfer experiments. When employing InceptionV3 as the sole attack model, SRAP attains the highest transfer attack success rates across all tested models except for ResNet50. Notably, it achieves attack success rates of 0.869 on AlexNet and 0.889 on MobileNetV3-small, demonstrating SRAP's potent attack efficacy even with a single surrogate model. The subpar performance of ResNet50 is attributed to the substantial structural differences between InceptionV3 and ResNet50. Additionally, while SRAP shows the best attack performance on Swin-tiny among its counterparts, this effect is still significantly lower than other CNN-based models.

The incorporation of ensemble models markedly enhances the transferability of adversarial examples. In SRAP, the three ensemble models are assigned equal weights. For RGAN, the ensemble is considered a singular model during generator training, with its collective output used for iterative parameter updates. A similar approach is employed in RAE-BIM. Given their inherent support for ensemble-based attacks, the original strategies outlined in their respective papers are adopted in the case of DP-RAE and BRAE. Experimental findings indicate that the ensemble model strategy substantially boosts adversarial transferability across various models. Notably, when targeting the ResNet50 model, including a structurally similar ResNet152 model within the ensemble increased the attack success rate from 0.535 to 0.968. Although this improvement is observed in other models, it is less pronounced.

Consequently, it is reasonable to hypothesize that ensemble models comprising a greater diversity of structures, such as a mix of CNN-based and ViT-based models, will generate adversarial examples with superior transferability. Nonetheless, our experiments reveal that the complexity of generating

adversarial examples increases as model diversity increases, making the loss function more challenging to converge and optimize. Therefore, a delicate balance between model diversity and the optimization challenge must be achieved. In ensemble model experiments, SRAP consistently performs well across different test models, achieving an average attack success rate of 0.806, which surpasses the state-of-the-art DP-RAE's rate of 0.576. This result underscores the effectiveness of SRAP's attack method in enhancing the transferability of reversible adversarial examples.

D. Robustness Evaluation

Image processing techniques to defend against adversarial examples and diminish their attack efficacy have been established as an efficacious defensive strategy. Consequently, when employing adversarial example techniques for data safeguarding, it is imperative to consider the adversarial examples' robustness. This study assesses the robustness of prevalent adversarial example defense methodologies and typical image processing techniques. Adhering to the testing protocol outlined in DP-RAE [7], we designated Random Resize and padding (Random) [44], Gaussian blurring (Gaussian) [45], JPEG compression (JPEG) [46], and super-resolution (Super) [47] as representative image processing algorithms to gauge the robustness of adversarial example defenses. All defense parameters were maintained following the original research papers.

Furthermore, we accounted for the influence of social media platforms. Images disseminated through social media undergo secondary compression, potentially rendering adversarial examples ineffective. To this end, we selected two exemplary social media dissemination channels: WeiBo and WeChat. In the context of WeChat, adversarial examples with perturbations were transmitted via the chat interface, and the attack success rate was subsequently assessed using the target model after receipt by the intended recipient. For WeiBo, adversarial examples were uploaded through personal

TABLE IV

EXPERIMENTS RESULTS OF ROBUSTNESS TESTS FOR REVERSIBLE ADVERSARIAL EXAMPLES. THE BEST RESULTS ARE EMPHASIZED IN BOLD

Defense Methods	Attack Methods				
	DP-RAE	RGAN	BRAE	RAE-BIM	SRAP (Ours)
None	0.986	0.992	0.907	0.974	0.990
JPEG (qp=90)	0.694	0.779	0.891	0.359	0.945
JPEG (qp=70)	0.657	0.700	0.700	0.201	0.909
JPEG (qp=50)	0.633	0.620	0.689	0.130	0.889
Gaussian	0.592	0.655	0.723	0.085	0.990
Super	0.823	0.684	0.796	0.396	0.948
Random	0.626	0.629	0.789	0.126	0.960
WeChat	0.654	0.621	0.698	0.138	0.907
WeiBo	0.627	0.643	0.716	0.108	0.964
Average	0.671	0.678	0.765	0.216	0.940

accounts, downloaded from public domains, and then evaluated for attack success rate on the target model. The empirical findings are delineated in Table IV.

The experimental results reveal that current adversarial example defense methods substantially affect the attack efficacy of reversible adversarial examples. Specifically, the DP-RAE method achieves an attack success rate of 0.986 on InceptionV3; however, this rate decreases to 0.657 after JPEG compression with a quality factor of 70, representing a significant reduction of 0.329. A similar performance decline is observed on social media platforms like WeiBo and WeChat, which is understandable given their use of comparable compression algorithms to manage image data and conserve bandwidth. Notably, the SRAP method exhibits enhanced robustness in practical applications, maintaining its original attack success rate even under Gaussian blur conditions, indicating that SRAP is resilient to such blurring effects. This resilience is attributed to SRAP's utilization of adversarial patches instead of adversarial noise perturbations. By applying consistent adversarial perturbations across different regions, SRAP possesses more outstanding attack capabilities and enhances the robustness of the generated adversarial examples. This optimization strategy results in adversarial examples less susceptible to common disruptions.

E. Reversibility Evaluation

The restorability of reversible adversarial examples is crucial for determining their usability by the recipient upon receipt. This ability can be assessed through two primary dimensions. Firstly, the accuracy of recognition by the target network for the restored adversarial examples can be quantified using the attack success rate. Secondly, the visual discrepancy between the restored adversarial examples and the original clean samples was evaluated comprehensively via PSNR and SSIM.

In this experimental section, we conduct a restoration test of reversible adversarial examples on the InceptionV3 target model, with the results presented in Table V. The findings indicate that all reversible adversarial examples exhibit satisfactory performance in terms of recognition restoration. Notably,

TABLE V

REVERSIBILITY EVALUATION RESULTS

Metrics	Attack Methods					
	DP-RAE	RGAN	BRAE	RAE-BIM	GEAA	SRAP (Ours)
ASR↓	0.000	0.006	0.000	0.000	0.007	0.000
PSNR↑	44.40	43.76	48.94	55.47	48.98	62.29
SSIM↑	0.9801	0.9548	0.9764	0.9960	0.9860	0.9998

only the samples restored using RGAN and GEAA methods induce slight misidentification by the target network. This phenomenon is attributed to the inherent limitations of denoiser or restorer structures in achieving complete restoration, as they essentially rely on information reconstruction. Conversely, reversible information-hiding techniques can achieve precise restoration, ensuring the target model accurately recognizes the restored samples.

Regarding visual effect restoration, SRAP performs better than other methods, as evidenced by higher PSNR and SSIM indicators. Despite also utilizing reversible information-hiding techniques, SRAP's enhanced visual outcomes can be attributed to the following factors. Firstly, while reversible information-hiding techniques minimally alter the original samples, the restoration process can encounter boundary issues, resulting in truncation and incomplete restoration. The extent of these effects increases with the amount of data hidden. SRAP, however, hides a relatively modest amount of information, approximately 500 bits, leading to even less modification of the carrier and better restoration results. Secondly, extensive modifications to the original image can introduce restoration losses during the generation of adversarial samples. For instance, the DP-RAE method applies adversarial attacks across the entire image, with varying perturbations at each pixel, complicating the restoration process. Conversely, SRAP modifies only 12.75% of the image area and incorporates quantization during the modification, thereby reducing the restoration complexity. In conclusion, SRAP's adoption of a small-size adversarial patch strategy and lower information capacity contributes to its superior restoration effects.

F. Commercial Model Attack

To test commercial image recognition APIs, we opted for Baidu Cloud's commercial image recognition API, which is recognized as one of the industry's most mature and advanced solutions for general image recognition. Baidu Cloud's API categorizes images at two levels: the primary level encompasses broad categories such as birds, fish, and cars, while the secondary level involves fine-grained categories, further subdividing the broad category of fish into species like crucian carp and goldfish.

We selected 100 SRAP adversarial sample images and processed them through Baidu Cloud's official API interface. Initially, we identified the correct classification labels for the clean samples. Subsequently, we recognized the adversarial samples paired with each clean sample. An adversarial sample is deemed to have successfully attacked both broad and fine-grained categories if the broad category differs. If the

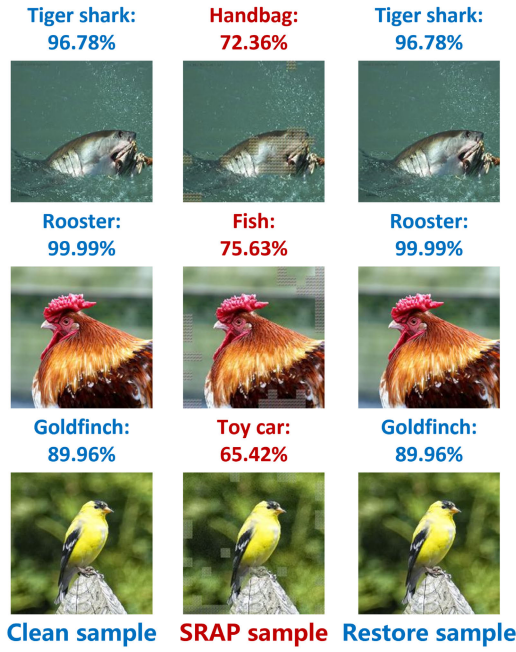


Fig. 5. Samples of SRAP adversarial attack, the numbers mean classification confidence.

TABLE VI
RESULTS OF THE ATTACK TEST ON BAIDU CLOUD API

Metrics	Broad Label	Fine-grained Label
Top-1 ASR	0.64	0.88
Top-5 ASR	0.58	0.78
Average Top-1 confidence	0.53	0.38

broad category remains the same but the fine-grained recognition varies, it is considered a successful attack only on the fine-grained category. If both categories match, the attack is deemed unsuccessful.

The results of the experimental tests, presented in Table VI, indicate an attack success rate of 0.64 for broad-category attacks and 0.88 for fine-grained recognition attacks. These findings demonstrate the effectiveness of SRAP adversarial samples in attacking commercial APIs. Fig. 6 visually represents the attack's efficacy against Baidu Cloud API.

G. Adaptive Protection

To counter potential proactive defenses in real-world applications, we acknowledge the risk of illegal intruders utilizing adversarial patch detection techniques to manipulate images and conceal malicious patches, thereby circumventing our defensive measures. In light of this threat, we conducted tests using two methods: Patch Detection [48] and Patch Zero [49]. We assessed the attack success rate under such conditions by zeroing out the detected regions. We selected adversarial samples generated by the InceptionV3 target model for our tests. Additionally, under a stricter scenario where the attacker visually inspects and manually removes all potential



Fig. 6. Samples of SRAP adversarial attack on **Baidu Cloud**'s commercial image recognition API.

TABLE VII
ADAPTIVE PROTECTION EXPERIMENTS

Detection Methods	ASR
Patch Detection	0.956
Patch Zero	0.944
Manual removal	0.790

adversarial patches, the ultimate experimental outcomes are documented in Table VII.

The table indicates that the efficacy of employing adversarial patch detection techniques and subsequently removing the patches is unsatisfactory. This outcome is attributed to the fact that existing detection methods are primarily tailored for identifying singular, large-scale adversarial patches. Consequently, they exhibit limited performance in detecting the small, numerous patches characteristic of SRAP, thereby hindering effective removal. Regarding manual removal, the adversarial patches were strategically placed in model-sensitive areas during their addition. Zeroing out these areas leads to the loss of significant valid information, ultimately resulting in recognition failure. In conclusion, SRAP demonstrates robust resistance against proactive attacker strategies.

Additionally, currently, adversarial training and adversarial defense have become relatively mature technologies. To test the accuracy of different reversible adversarial attack methods against these robustness-enhanced classification models, we selected two robust models from the open-source RobustBench [52] platform. These models are Liu2023Comprehensive_Swin-L [50] and Bai2024MixedNUTS [51]. For the ImageNet dataset, these two models—based on the Swin-L model—achieve robust accuracies of 59.56% and 58.40% respectively under RobustBench's Linf, eps= 4/255 test benchmark. The experimental results are presented in Table VIII.

TABLE VIII
EXPERIMENTS ON ROBUST MODELS

Target Models	Attack Methods	ASR
Liu2023 [50] Comprehensive_Swin-L	DP-RAE	0.736
	BRAE	0.806
	RAE-BIM	0.796
	RGAN	0.733
	GEAA	0.644
	SRAP (Ours)	0.844
Bai2024 [51] MixedNUTS	DP-RAE	0.757
	BRAE	0.774
	RAE-BIM	0.809
	RGAN	0.692
	GEAA	0.665
	SRAP (Ours)	0.848

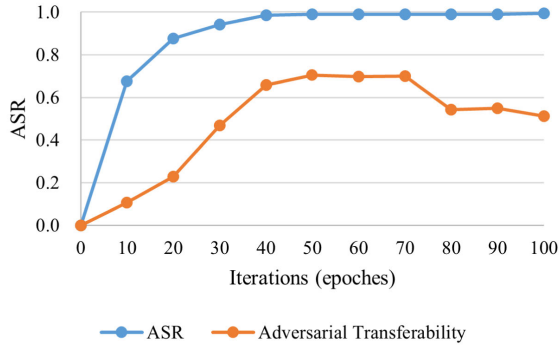


Fig. 7. An investigation was conducted to assess the influence of iteration cycles on the success rate of SRAP attacks and their adversarial transferability, utilizing the InceptionV3 model as the target model.

From the experimental results, it can be observed that the attack success rates of different reversible adversarial attack methods on robust models have decreased to a certain extent compared with models without robustness enhancement. The decrease range is between 0.1 and 0.2. Among these attack methods, the SRAP method exhibits the best attack ability, achieving attack success rates of 0.844 and 0.848 respectively. This data is significantly higher than the attack success rates in RobustBench evaluations, and we attribute this to the following reasons. First, we used open-source model parameters. These robust models have not undergone specialized training on reversible adversarial examples, so their generalization ability for this unknown attack method is reduced. Second, during the reversible adversarial attack, a larger attack magnitude is used, along with the designed and more intelligent selection of attack perturbation regions. Thus, such a high attack success rate can be achieved.

H. Ablation Study

In the ablation experiment section, we investigated the impact of iteration cycles and adversarial patch size on the effectiveness of SRAP attacks.

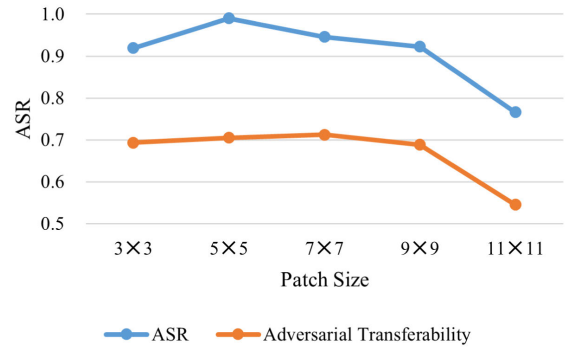


Fig. 8. The effect of adversarial patch size on the efficacy of adversarial attacks, utilizing the InceptionV3 model as our target. All added patches were consistently set to cover an area of 12.75%.

Firstly, regarding the influence of iteration cycles on attack success rate, as shown in Fig. 7, it can be observed that as the number of iteration cycles increases, the attack success rate generally trends upwards, and the transferability of adversarial attacks also exhibits an increasing trend. However, when the iteration cycles reach 50, the improvement in attack success rate becomes minimal, and when the iteration cycles exceed 70, the transferability of adversarial samples starts to decline. We believe this is because, with the increase in iteration cycles, adversarial samples may experience overfitting, meaning they only exhibit strong attack effects on the target model, while their attack effectiveness on other models decreases. Therefore, in selecting the number of iteration cycles, we chose 50 to balance the attack success rate and adversarial transferability.

Fig. 8 presents the impact of varying the minimum adversarial patch size on SRAP effectiveness, the attack success rate peaks at a patch size of 5×5 . While a larger patch initially boosts SRAP's adversarial transferability, further enlargement reduces this transferability. The underlying reasons, we hypothesize, are twofold. Firstly, an increase in patch size from 3×3 to 5×5 allows for the inclusion of more adversarial information, thereby enhancing the adversarial capacity and, consequently, the attack success rate. However, as the patch size grows, accommodating this additional information within the adversarial sample's steganographic capacity necessitates increasing the quantization step size. Furthermore, a larger patch size restricts the flexibility in selecting optimal positions for patch placement, ultimately decreasing the efficacy of adversarial attacks. Therefore, in SRAP, a balance between these factors leads us to opt for a 5×5 patch size for adversarial modifications.

The results of ablation experiments on the mask selection strategy and the effectiveness of adversarial noise for the SRAP method are presented in Table IX. Gradient Mask and Adversarial Noise are the strategies adopted in this paper. Random Mask refers to randomly selecting masks of the same quantity from an image for adversarial patch addition. Gaussian refers to adding Gaussian random noise to the selected masks, with the addition magnitude and quantization step size consistent with those of adversarial noise. It can be observed that the Gradient Mask strategy proposed in this paper can

TABLE IX
ABLATION STUDY ON MASK STRATEGY AND NOISE TYPE

Mask Strategy		Noise Type		ASR
Gradient Mask	Random Mask	Adversarial	Gaussian	
	✓		✓	0.102
✓			✓	0.199
	✓	✓		0.794
✓		✓		0.990

effectively improve the attack success rate of adversarial noise. Compared with Random Mask, the attack success rate increases by 0.196. Moreover, adversarial noise also shows a significant improvement in attack effectiveness compared with random Gaussian noise, reaching a level of 0.791. This fully demonstrates the effectiveness of the strategies in the SRAP method proposed in this paper.

V. CONCLUSION

Existing data protection approaches utilizing reversible adversarial example techniques must pay more attention to robustness and adversarial transferability. Our paper introduces a novel technique for generating reversible adversarial examples, employing self-reversible adversarial patches (SRAP). This method involves creating small adversarial patches and strategically placing multiple non-overlapping patches in key locations. This results in adversarial examples that exhibit high attack success rates, robustness, and adversarial transferability. Experimental results across various target models confirm the superior adversarial transferability and robustness of our proposed SRAP technique. Future work will focus on reducing these reversible adversarial attacks' visual prominence while enhancing their transferability.

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